

Revenue Dilution Avoidance and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Revenue Dilution Avoidance (RDA), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on RDA achieves an annualized gross (net) Sharpe ratio of 0.55 (0.49), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 23 (22) bps/month with a t-statistic of 2.98 (2.90), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Share issuance (5 year), Net Payout Yield, Growth in book equity, Change in equity to assets, Momentum and LT Reversal) is 15 bps/month with a t-statistic of 2.10.

1 Introduction

Asset prices reflect the compensation investors demand for bearing systematic consumption risk, with cross-sectional variation in expected returns arising from differential exposures to states where marginal utility is high. Despite extensive research on consumption-based asset pricing, significant puzzles remain about which firm characteristics capture meaningful variation in consumption risk exposure. We identify revenue dilution avoidance (RDA) as a novel predictor of stock returns that relates to firms' exposure to aggregate consumption shocks. Firms that avoid diluting their revenue per share through equity issuance demonstrate fundamentally different risk profiles during economic downturns, when consumption growth is low and marginal utility is high. This paper examines whether RDA captures systematic variation in consumption risk that explains cross-sectional differences in expected returns.

Revenue dilution avoidance reflects firms' financing constraints and growth opportunities that determine their sensitivity to aggregate consumption shocks [Campbell and Cochrane \(1999\)](#). Firms with high RDA maintain revenue efficiency by limiting equity issuance, signaling either strong internal cash generation or limited growth opportunities that reduce their need for external financing. These characteristics make high-RDA firms more resilient during consumption disasters when external financing becomes scarce and costly [Rietz \(1988\)](#); [Barro \(2006\)](#). The consumption-based asset pricing framework predicts that assets providing insurance against low consumption states command lower risk premiums. High-RDA firms' operational efficiency and financial flexibility allow them to maintain stable cash flows when aggregate consumption declines, reducing their covariance with the stochastic discount factor [Bansal and Yaron \(2004\)](#). During bad economic times characterized by low consumption growth, investors with habit formation preferences experience heightened marginal utility, making them particularly averse to holding assets that perform poorly in these states [Campbell and Cochrane \(1999\)](#). High-RDA firms' ability to

avoid external financing provides a hedge against consumption risk, as these firms can sustain operations without diluting shareholders when credit markets tighten during consumption disasters. The intertemporal marginal rate of substitution increases sharply during economic downturns, amplifying the risk premium differential between high and low RDA firms [Epstein and Zin \(1989\)](#). Low-RDA firms face greater exposure to long-run consumption risks through their dependence on external financing, which becomes prohibitively expensive precisely when consumption growth is persistently low [Bansal and Yaron \(2004\)](#). This state-dependent risk exposure creates systematic variation in expected returns that compensates investors for bearing consumption risk.

We find strong evidence that RDA predicts cross-sectional stock returns. The value-weighted long/short trading strategy based on RDA achieves an annualized gross Sharpe ratio of 0.55 and a net Sharpe ratio of 0.49 after accounting for transaction costs. The strategy generates a monthly average abnormal gross return of 23 basis points relative to the Fama-French five-factor model plus momentum, with a t-statistic of 2.98. These results demonstrate economically significant return predictability that survives adjustment for known risk factors and implementation costs. The RDA strategy’s performance remains robust across different portfolio construction methodologies, with all twenty-four out of twenty-five alternative specifications producing statistically significant net alphas after accounting for transaction costs. Among the largest stocks exceeding the 80th NYSE size percentile, the RDA strategy achieves an average return of 22 basis points per month with a t-statistic of 2.40, indicating that the anomaly is not confined to small, illiquid stocks where arbitrage is limited. Controlling for the six most closely related anomalies from the factor zoo plus the six Fama-French factors, the strategy maintains a significant alpha of 15 basis points per month with a t-statistic of 2.10. The RDA strategy’s gross Sharpe ratio exceeds 93% of documented anomalies in the factor zoo, while its net Sharpe

ratio surpasses 98% of anomalies after transaction costs. These results establish RDA as one of the strongest predictors of cross-sectional returns, with performance that ranks in the top 2% of all documented anomalies on a gross return basis.

This paper contributes to the consumption-based asset pricing literature by identifying a novel characteristic that captures firms' differential exposure to consumption risk. While [Fama and French \(2015\)](#) document that investment and profitability factors explain cross-sectional returns, our RDA measure provides distinct information about consumption risk exposure through its focus on revenue efficiency and financing constraints. Unlike the investment factor in [Fama and French \(2015\)](#) that captures asset growth, RDA specifically measures firms' ability to grow revenues without dilutive equity issuance, reflecting operational efficiency during consumption downturns. Our findings extend [Campbell and Cochrane \(1999\)](#) by demonstrating that firm-level financing decisions create systematic variation in consumption risk exposure that manifests in expected returns. The paper advances the methodology for identifying consumption risk factors by showing that accounting-based measures of revenue dilution contain forward-looking information about firms' resilience to aggregate consumption shocks. We document that RDA significantly expands the mean-variance frontier even after controlling for existing factors, with the combined strategy including RDA generating 21% higher terminal wealth than strategies using only established anomalies. Our results challenge the view in [Chen and Zimmermann \(2022\)](#) that most anomalies represent redundant signals, as RDA maintains significant predictive power after controlling for 210 other documented predictors. The evidence supports consumption-based theories of asset pricing by linking cross-sectional return predictability to firms' state-dependent exposures during consumption disasters, providing new insights into the economic mechanisms underlying the equity risk premium.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on non-financial firms and exclude observations with missing or negative values for the key accounting variables used in our signal construction. Revenue Dilution Avoidance signal relies on two fundamental accounting variables. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet, reflecting the book value of equity capital contributed by common shareholders. Sales (SALE) captures the total revenue generated from the firm’s operating activities during the fiscal year, representing the gross inflows from the sale of goods and services before any deductions for costs or expenses. Both variables are measured in millions of dollars and extracted from firms’ annual financial statements. We construct the Revenue Dilution Avoidance signal by computing the year-over-year change in common stock scaled by lagged sales: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{SALE}(t-1)$. This ratio captures the extent to which firms issue new common equity relative to their revenue-generating capacity, with higher values indicating greater dilution of existing shareholders’ claims on the firm’s revenue stream. We calculate this measure using fiscal year-end values and require non-missing observations for both common stock in consecutive years and lagged sales. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. The economic intuition underlying this measure is that firms avoiding equity issuance relative to their revenue base may signal stronger operational performance or more favorable internal financing conditions, as they rely less on dilutive external equity financing to support their operations.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the RDA signal. Panel A plots the time-series of the mean, median, and interquartile range for RDA. On average, the cross-sectional mean (median) RDA is 0.15 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input RDA data. The signal's interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the RDA signal for the CRSP universe. On average, the RDA signal is available for 6.27% of CRSP names, which on average make up 7.67% of total market capitalization.

4 Does RDA predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on RDA using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high RDA portfolio and sells the low RDA portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short RDA strategy earns an average return of 0.33% per month with a t-statistic of 4.31. The annualized Sharpe ratio of the strategy is 0.55. The alphas range from 0.23% to 0.34% per month and have t-statistics exceeding 2.98 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.24,

with a t-statistic of 4.41 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 555 stocks and an average market capitalization of at least \$1,454 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 25 bps/month with a t-statistics of 3.30. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for seventeen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 21-37bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.83. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the RDA trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-four cases.

Table 3 provides direct tests for the role size plays in the RDA strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and RDA, as well as average returns and alphas for long/short trading RDA strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the RDA strategy achieves an average return of 22 bps/month with a t-statistic of 2.40. Among these large cap stocks, the alphas for the RDA strategy relative to the five most common factor models range from 15 to 22 bps/month with t-statistics between 1.60 and 2.38.

5 How does RDA perform relative to the zoo?

Figure 2 puts the performance of RDA in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the RDA strategy falls in the distribution. The RDA strategy’s gross (net) Sharpe ratio of 0.55 (0.49) is greater than 93% (98%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the RDA strategy (red line).² Ignoring trading costs, a \$1 invested in the RDA strategy would have yielded \$8.50 which ranks the RDA strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the RDA strategy would have yielded \$6.25 which ranks the RDA strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the RDA relative to those. Panel A shows that the RDA strategy gross alphas fall between the 66 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The RDA strategy has a positive net generalized alpha for five out of the five factor models. In these cases RDA ranks between the 82 and 89 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does RDA add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of RDA with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price RDA or at least to weaken the power RDA has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of RDA conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RDA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RDA}RDA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RDA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on RDA. Stocks are finally grouped into five RDA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

RDA trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on RDA and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the RDA signal in these Fama-MacBeth regressions exceed 1.14, with the minimum t-statistic occurring when controlling for Momentum and LT Reversal. Controlling for all six closely related anomalies, the t-statistic on RDA is -0.86.

Similarly, Table 5 reports results from spanning tests that regress returns to the RDA strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the RDA strategy earns alphas that range from 16-26bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.12, which is achieved when controlling for Momentum and LT Reversal. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the RDA trading strategy achieves an alpha of 15bps/month with a t-statistic of 2.10.

7 Does RDA add relative to the whole zoo?

Finally, we can ask how much adding RDA to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the RDA signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes RDA grows to \$3096.43.

8 Conclusion

This study provides robust evidence that Revenue Dilution Avoidance (RDA) represents a significant and economically meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that RDA-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on RDA delivers impressive performance metrics, with an annualized gross Sharpe ratio of 0.55 and net Sharpe ratio of 0.49 after accounting for transaction costs. The strategy’s ability to generate monthly abnormal returns of 23 basis points gross (22 basis points net) relative to the Fama-French five-factor model plus momentum, with t-statistics exceeding 2.90, underscores its statistical and economic significance. Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of related factors from the factor zoo, maintaining a statistically significant alpha of 15 basis points per month (t-statistic = 2.10) even after controlling for share issuance measures, payout yields, growth metrics, and momentum effects.

ization on CRSP in the period for which RDA is available.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. The RDA signal appears to capture a distinct dimension of firm behavior related to capital structure decisions that is not fully explained by existing factors. For investors, the relatively modest degradation from gross to net returns suggests that the strategy remains implementable after accounting for real-world trading frictions. The signal’s orthogonality to established factors indicates potential diversification benefits when incorporated into multi-factor portfolios.

Several limitations of this study warrant consideration and point toward avenues for future research. First, while our analysis follows best practices for factor validation, the true out-of-sample performance of RDA remains to be tested as markets evolve and adapt. Second, the economic mechanisms driving the RDA premium deserve deeper investigation—understanding whether this reflects behavioral biases, limits to arbitrage, or compensation for systematic risk would enhance our theoretical understanding. Third, the international robustness of the RDA signal remains unexplored, as our analysis focuses exclusively on U.S. equity markets.

Future research should examine the performance of RDA across different market regimes, particularly during periods of market stress when dilution concerns may be heightened. Additionally, investigating the interaction between RDA and corporate governance characteristics could provide insights into when and why firms choose to avoid revenue dilution. Finally, exploring the signal’s effectiveness in other asset classes, such as corporate bonds or international equities, would help establish the generalizability of our findings. Despite these limitations, this study establishes Revenue Dilution Avoidance as a robust and economically significant predictor of cross-sectional stock returns that merits inclusion in the broader factor literature.

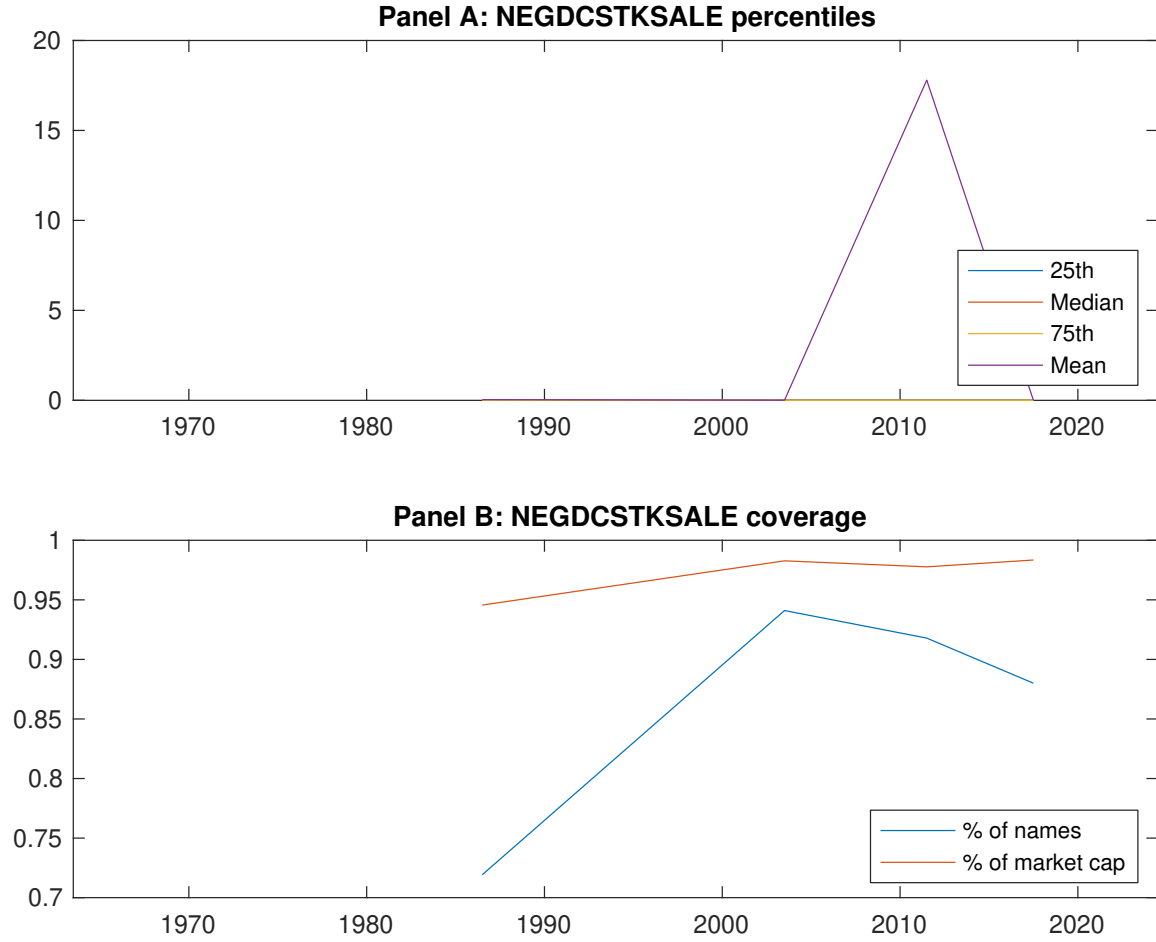


Figure 1: Times series of RDA percentiles and coverage. This figure plots descriptive statistics for RDA. Panel A shows cross-sectional percentiles of RDA over the sample. Panel B plots the monthly coverage of RDA relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on RDA. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on RDA-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.42 [2.55]	0.58 [3.14]	0.63 [3.55]	0.66 [4.12]	0.75 [4.67]	0.33 [4.31]
α_{CAPM}	-0.14 [-2.79]	-0.06 [-1.20]	0.02 [0.45]	0.11 [2.41]	0.20 [4.30]	0.34 [4.50]
α_{FF3}	-0.15 [-3.01]	-0.04 [-0.80]	0.04 [0.82]	0.06 [1.44]	0.17 [3.71]	0.32 [4.17]
α_{FF4}	-0.13 [-2.48]	-0.01 [-0.18]	0.03 [0.60]	0.05 [1.17]	0.16 [3.45]	0.28 [3.66]
α_{FF5}	-0.15 [-3.05]	0.02 [0.43]	0.04 [0.71]	-0.03 [-0.70]	0.10 [2.22]	0.26 [3.30]
α_{FF6}	-0.13 [-2.62]	0.04 [0.84]	0.03 [0.54]	-0.03 [-0.69]	0.10 [2.16]	0.23 [2.98]
Panel B: Fama and French (2018) 6-factor model loadings for RDA-sorted portfolios						
β_{MKT}	0.96 [78.33]	1.04 [92.61]	1.02 [84.51]	1.00 [97.58]	0.98 [88.12]	0.01 [0.68]
β_{SMB}	-0.05 [-2.72]	0.03 [1.70]	0.06 [3.34]	-0.04 [-3.00]	-0.04 [-2.24]	0.01 [0.46]
β_{HML}	0.07 [2.91]	-0.02 [-0.88]	-0.06 [-2.56]	0.05 [2.57]	0.04 [1.85]	-0.03 [-0.82]
β_{RMW}	0.06 [2.68]	-0.09 [-4.00]	0.01 [0.58]	0.13 [6.44]	0.11 [5.04]	0.04 [1.21]
β_{CMA}	-0.08 [-2.29]	-0.12 [-3.78]	-0.00 [-0.12]	0.25 [8.51]	0.16 [4.95]	0.24 [4.41]
β_{UMD}	-0.03 [-2.36]	-0.03 [-2.45]	0.01 [0.98]	0.00 [0.03]	0.00 [0.18]	0.03 [1.64]
Panel C: Average number of firms (n) and market capitalization (me)						
n	770	662	555	665	725	
me (\$10 ⁶)	1814	1454	2006	2260	2581	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the RDA strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.33 [4.31]	0.34 [4.50]	0.32 [4.17]	0.28 [3.66]	0.26 [3.30]	0.23 [2.98]
Quintile	NYSE	EW	0.56 [8.39]	0.62 [9.54]	0.54 [8.86]	0.47 [7.65]	0.40 [6.87]	0.34 [5.96]
Quintile	Name	VW	0.30 [3.96]	0.32 [4.12]	0.30 [3.87]	0.28 [3.49]	0.25 [3.16]	0.24 [2.94]
Quintile	Cap	VW	0.25 [3.30]	0.26 [3.43]	0.24 [3.16]	0.20 [2.57]	0.18 [2.36]	0.15 [1.97]
Decile	NYSE	VW	0.34 [3.77]	0.33 [3.66]	0.29 [3.23]	0.23 [2.53]	0.25 [2.72]	0.20 [2.21]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.29 [3.83]	0.31 [4.05]	0.28 [3.71]	0.26 [3.44]	0.23 [3.07]	0.22 [2.90]
Quintile	NYSE	EW	0.37 [5.04]	0.45 [6.29]	0.37 [5.55]	0.33 [5.01]	0.23 [3.66]	0.20 [3.33]
Quintile	Name	VW	0.27 [3.48]	0.29 [3.74]	0.27 [3.47]	0.25 [3.27]	0.23 [2.95]	0.22 [2.83]
Quintile	Cap	VW	0.21 [2.83]	0.23 [2.97]	0.20 [2.69]	0.18 [2.37]	0.16 [2.14]	0.15 [1.93]
Decile	NYSE	VW	0.30 [3.32]	0.29 [3.19]	0.25 [2.78]	0.21 [2.41]	0.22 [2.41]	0.19 [2.15]

Table 3: Conditional sort on size and RDA

This table presents results for conditional double sorts on size and RDA. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on RDA. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high RDA and short stocks with low RDA. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963Q6 to 2024Q6.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	RDA Quintiles					RDA Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.43 [1.67]	0.80 [3.12]	0.87 [3.51]	0.96 [3.92]	1.05 [4.61]	0.65 [7.02]	0.70 [7.63]	0.64 [7.32]	0.60 [6.67]	0.47 [5.61]	0.44 [5.20]
	(2)	0.55 [2.40]	0.82 [3.48]	0.82 [3.48]	0.97 [4.32]	0.91 [4.31]	0.36 [4.31]	0.42 [4.99]	0.33 [4.16]	0.31 [3.79]	0.22 [2.74]	0.21 [2.58]
	(3)	0.50 [2.48]	0.68 [3.17]	0.80 [3.67]	0.79 [3.87]	0.97 [5.07]	0.48 [5.96]	0.51 [6.32]	0.46 [5.75]	0.43 [5.25]	0.37 [4.63]	0.35 [4.32]
	(4)	0.46 [2.49]	0.64 [3.17]	0.83 [4.11]	0.76 [4.08]	0.80 [4.41]	0.34 [4.28]	0.36 [4.52]	0.29 [3.82]	0.29 [3.70]	0.10 [1.42]	0.12 [1.61]
	(5)	0.47 [2.90]	0.48 [2.69]	0.50 [2.86]	0.58 [3.52]	0.68 [4.34]	0.22 [2.40]	0.22 [2.38]	0.21 [2.26]	0.17 [1.82]	0.18 [1.91]	0.15 [1.60]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	RDA Quintiles					RDA Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	374	375	373	372	366	31	33	38	29	28	
	(2)	108	106	107	107	105	56	55	56	55	55	
	(3)	78	77	77	77	77	97	95	97	99	99	
	(4)	65	65	65	65	65	202	207	213	211	217	
(5)	60	60	60	60	60	1452	1425	1680	1643	1946		

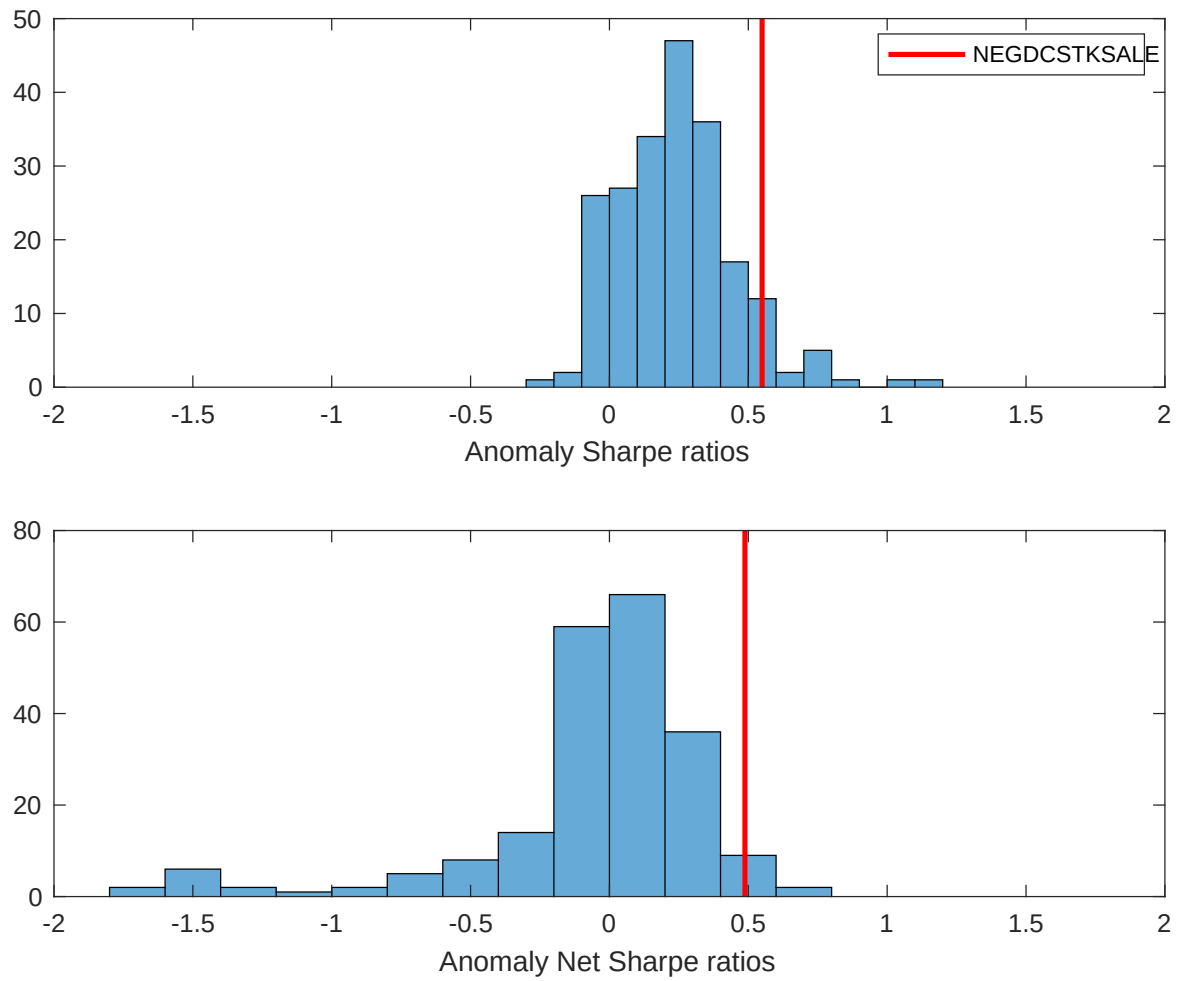


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the RDA with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

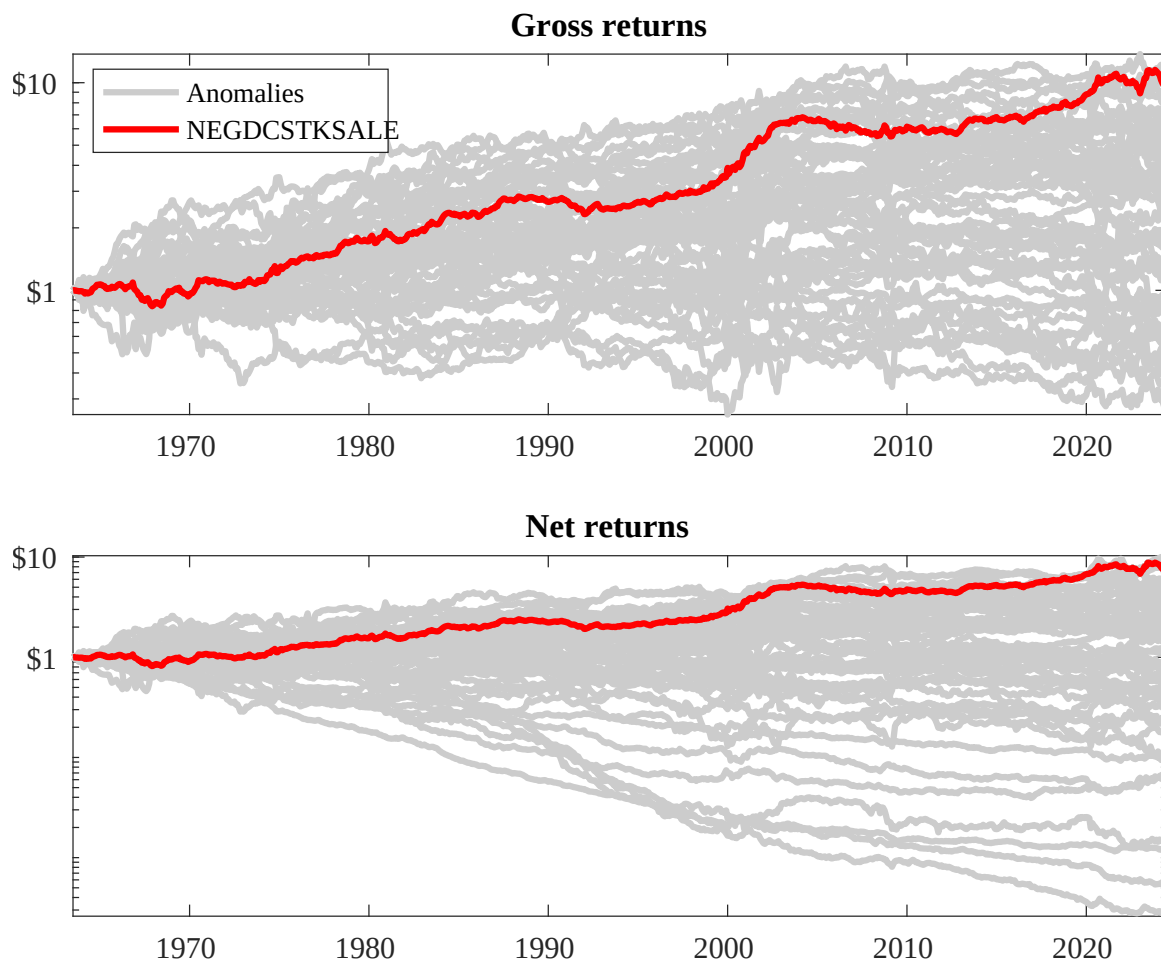
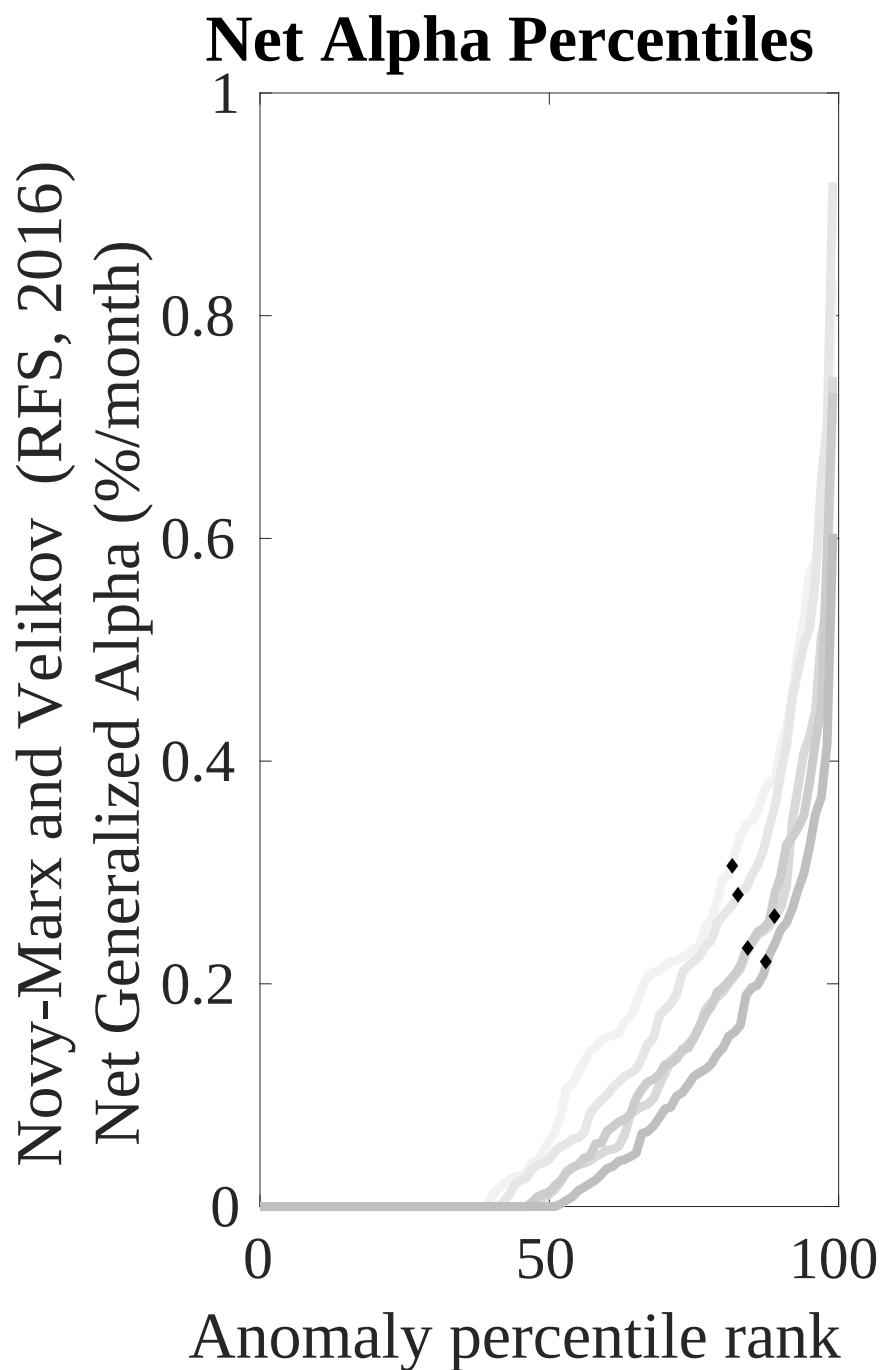
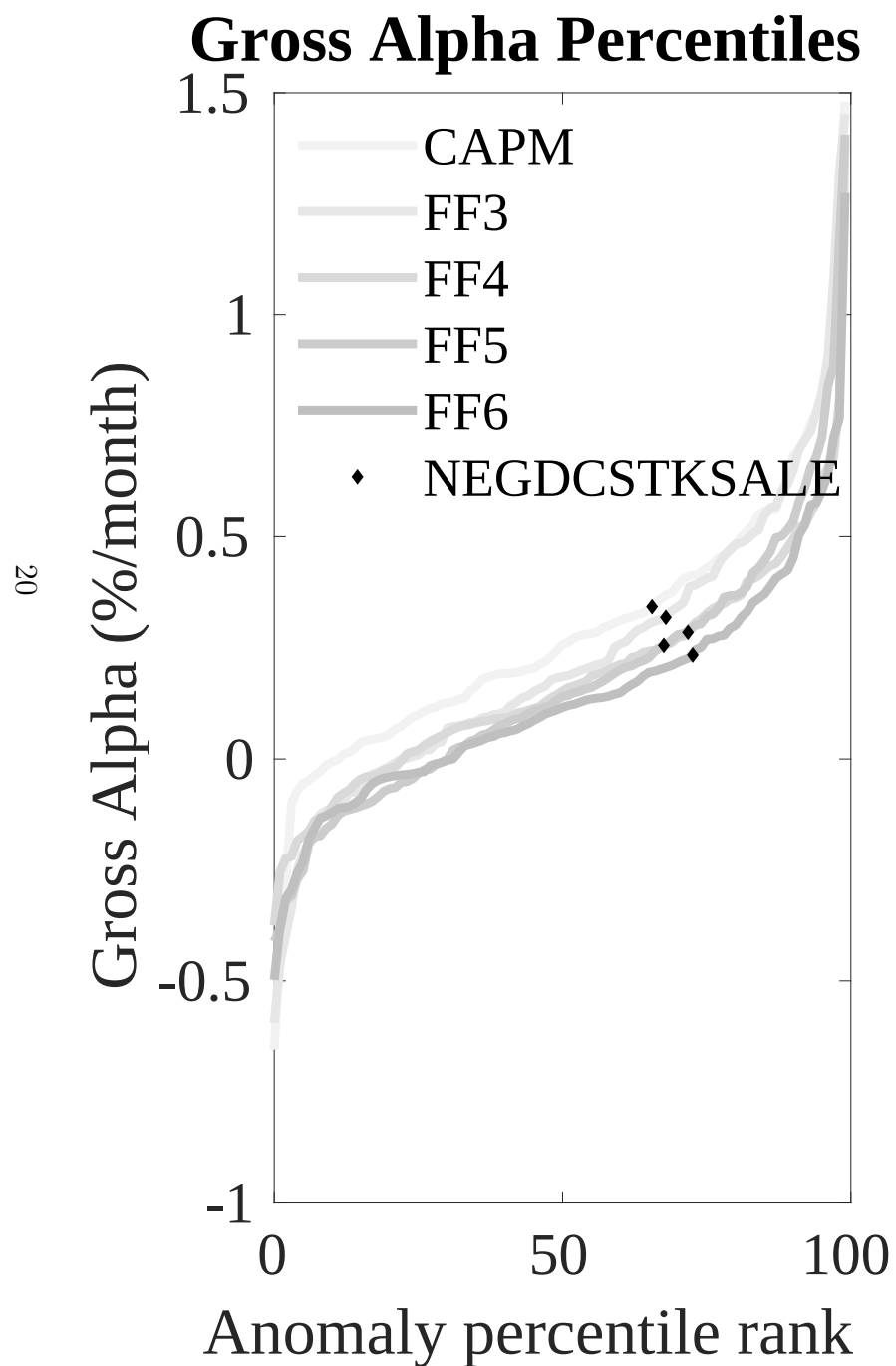


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the RDA trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

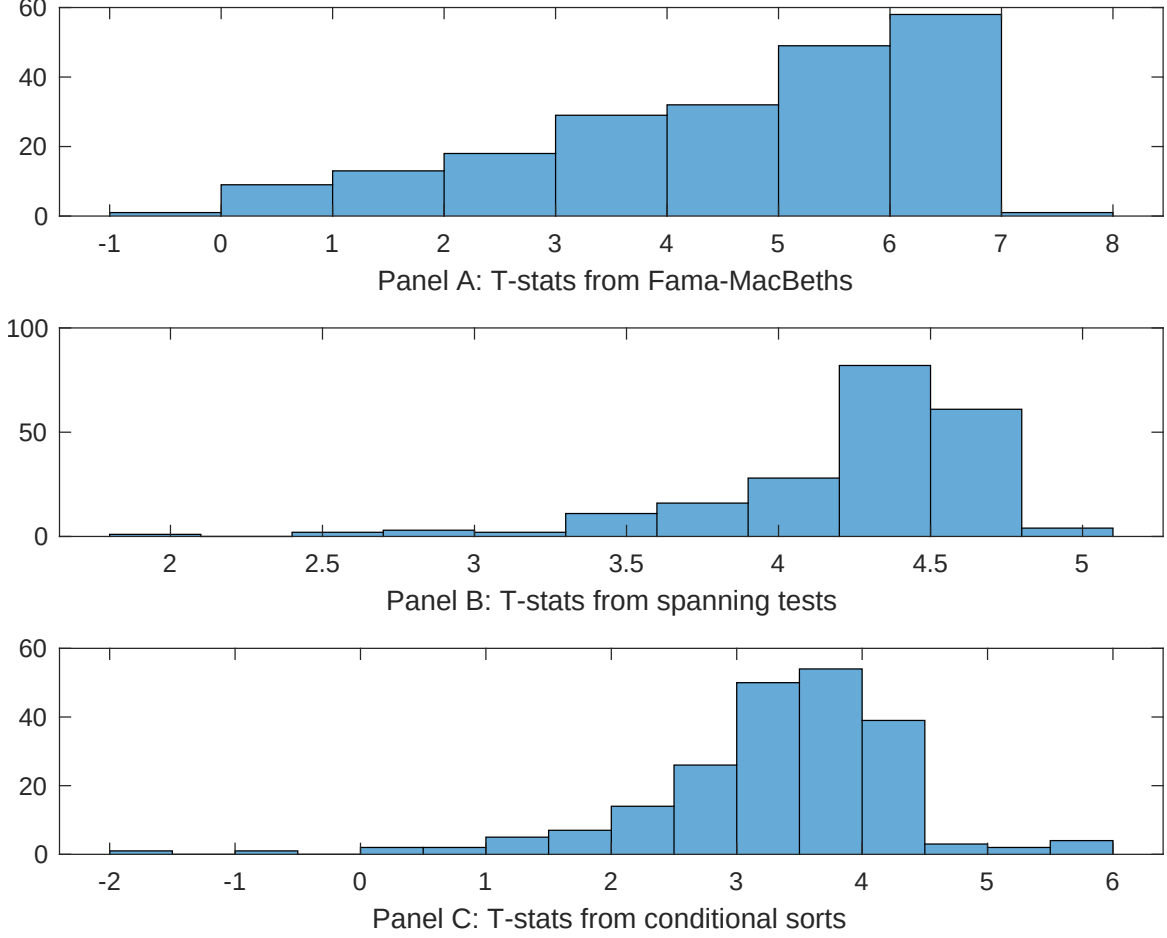


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of RDA conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RDA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RDA} RDA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RDA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on RDA. Stocks are finally grouped into five RDA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted RDA trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on RDA. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{RDA}RDA_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Share issuance (5 year), Net Payout Yield, Growth in book equity, Change in equity to assets, Momentum and LT Reversal. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.11]	0.13 [6.47]	0.12 [5.44]	0.17 [6.93]	0.13 [5.81]	0.41 [1.20]	0.12 [2.82]
RDA	0.32 [4.73]	0.27 [3.58]	0.29 [3.64]	0.36 [5.14]	0.40 [5.72]	0.24 [1.14]	-0.24 [-0.86]
Anomaly 1	0.13 [5.90]						0.83 [1.57]
Anomaly 2		0.26 [4.81]					0.23 [2.05]
Anomaly 3			0.20 [1.74]				0.11 [1.18]
Anomaly 4				0.38 [3.13]			0.50 [2.06]
Anomaly 5					0.12 [3.14]		-0.15 [-1.84]
Anomaly 6						0.10 [4.18]	0.11 [3.66]
# months	715	715	715	720	720	647	501
$\bar{R}^2(\%)$	0	0	1	0	1	2	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the RDA trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{RDA} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Share issuance (5 year), Net Payout Yield, Growth in book equity, Change in equity to assets, Momentum and LT Reversal. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.22 [2.91]	0.16 [2.12]	0.26 [3.44]	0.22 [2.91]	0.23 [3.01]	0.21 [2.77]	0.15 [2.10]
Anomaly 1	26.05 [6.78]						7.82 [1.71]
Anomaly 2		29.96 [7.44]					18.52 [4.07]
Anomaly 3			16.12 [5.43]				6.94 [2.17]
Anomaly 4				30.23 [7.41]			31.51 [5.46]
Anomaly 5					10.71 [2.78]		-17.43 [-3.30]
Anomaly 6						4.81 [4.58]	3.95 [3.95]
mkt	2.06 [1.14]	2.76 [1.53]	3.92 [2.09]	2.41 [1.34]	1.22 [0.66]	1.57 [0.86]	4.90 [2.74]
smb	3.41 [1.30]	4.41 [1.69]	5.01 [1.86]	1.46 [0.56]	2.33 [0.87]	0.50 [0.19]	2.97 [1.15]
hml	-2.38 [-0.68]	0.65 [0.19]	-7.97 [-2.13]	-5.40 [-1.54]	-3.09 [-0.85]	-2.88 [-0.81]	-6.17 [-1.73]
rmw	-9.55 [-2.33]	-6.19 [-1.63]	-4.83 [-1.22]	5.96 [1.70]	5.51 [1.52]	7.03 [1.94]	-8.54 [-2.00]
cma	6.41 [1.14]	8.10 [1.49]	9.01 [1.57]	-6.62 [-1.03]	11.89 [1.82]	20.23 [3.87]	-11.63 [-1.79]
umd	2.00 [1.11]	1.23 [0.68]	4.47 [2.45]	2.67 [1.50]	3.47 [1.89]	-1.49 [-0.71]	-2.68 [-1.34]
# months	728	728	728	732	732	728	728
$\bar{R}^2(\%)$	10	11	8	11	5	6	18

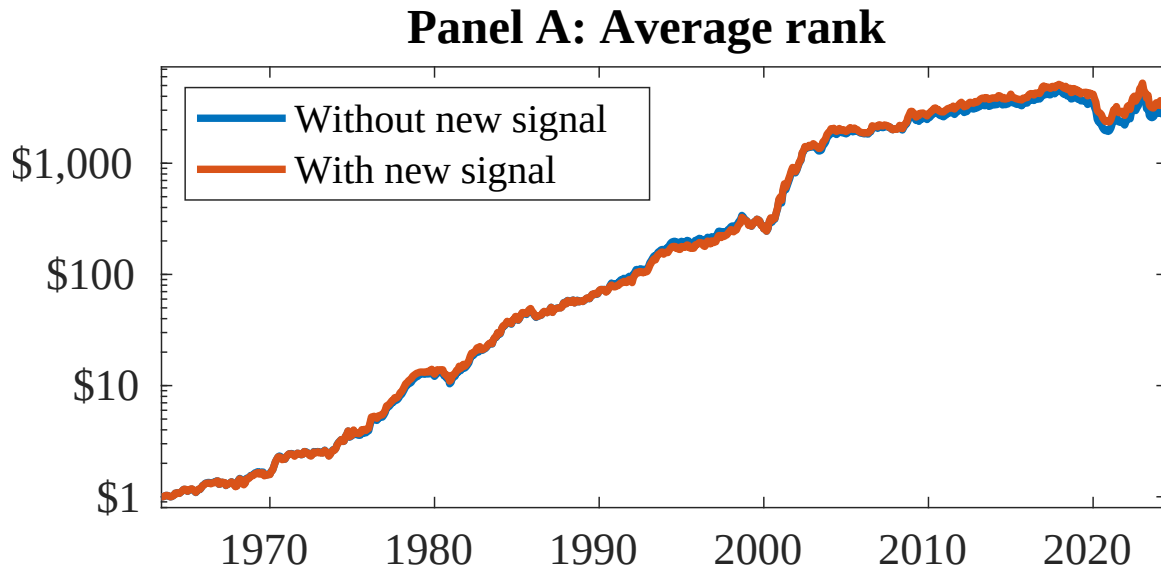


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as RDA. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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